

MATH0119 Dynamical Systems

<i>Year:</i>	2025–2026
<i>Code:</i>	MATH0119
<i>Level:</i>	7 (UG)
<i>Normal student group(s):</i>	UG: Year 3 or Year 4 Mathematics degrees PG: (2026-27) MSc Mathematics (level 7)
<i>Value:</i>	15 credits (= 7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	90% examination, 10% coursework
<i>Normal Pre-requisites:</i>	MATH0017 and (MATH0019 or MATH0020)
<i>Lecturer:</i>	Dr S Ghazouani

Course Description and Objectives

In this course, we will give an introduction to the theory of dynamical systems. A dynamical system is given by a transformation of a topological space which we think of as the law driving the evolution of a system over time (such as the evolution over time of the solar system, a population, the weather, you-name-it). Our goal will be to develop the tools and the language which allow for the general treatment of such problems: this will involve a good dose of topology, analysis and measure-theory.

Recommended Texts

1. Introduction to the Modern Theory of Dynamical Systems, by Katok and Hasselblatt
2. Introduction to Dynamical Systems, by Brin and Stuck
3. Ergodic theory: with a view towards number theory, by Einsiedler and Ward

Detailed Syllabus

- **Quasi-periodic dynamics.** Linear flows on the torus, circle homeomorphisms, rotation number, Poincaré theorem.
- **Chaotic dynamics.** Doubling map, shift maps, Smale's horseshoe
- **Hyperbolic dynamics.** Hyperbolic fixed points in $\mathbb{R}^2$, Stable Curve Theorem, Local Stability, homoclinic intersections
- **Ergodic theory.** Invariant measures, Ergodicity, Birkhoff's Ergodic Theorem